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HEALTH SAFETY & ENVIRONMENT PLAN

&

QUALITY POLICY





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1. INTRODUCTION

Safety in Construction Management deserves utmost attention especially in the hydrocarbon industry, such as Refineries, Pipelines and Gas Processing units etc. Construction is widely recognized as one of the accident-prone activities. Most of the accidents are caused by inadequate planning, failure during the construction process and/or because of design deficiencies. Besides property loss, accidents also result in injuries and fatalities to the personnel; same needs to be prevented.

The reasons for accidents during construction activities are related to unique nature of the industry, human behavior, difficult work-site conditions, extended odd duty hours, lack of training & awareness and inadequate safety management. Unsafe working methods, equipment failure and improper housekeeping also tend to increase the accident rate in construction.

Ensuring good quality of materials, equipment and competent supervision along with compliance of standard engineering practices shall go a long way to in built safety in the system.

The objective of this standard is to provide practical guidance on technical and educational framework for safety and health in construction with a view to:

- Prevent accidents and harmful effects on the health of workers arising from Employment in construction
- Ensure appropriate safety during implementation of construction;
- Provide safety practice guidelines for appropriate measures of planning, control and enforcement.

2. SCOPE

This document specifies broad guidelines on safe practices to be adhered to during construction activities in oil industry. However, before commencing any job, specific hazards and its effects should be assessed and necessary corrective/preventive actions should be taken by all concerned. The document is intended only to supplement and not to replace or supersede the prevailing statutory requirements, which shall also be followed as applicable. The scope of this document does not include the design aspects and quality checks during construction.

3. HSE POLICY



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ANS Constructions Private Limited

HEALTH,SAFETY & ENVIRONMENT (HSE) POLICY

ANS CONSTRUCTIONS PVT.LTD. (ANSCL) is committed to comply with the requirement of HSE Management System & continually improve its effectiveness by providing the appropriate processes, resources , principles, training & work culture.

In this endeavor, ANSCPL shall strive to be conscious of its responsibility towards creating , conserving & ascertaining a clean environment for sustainable development and protect the health & safety of all its associated workforce , ensuring thereby a safe , healthy & accident free work place.

To meet this commitment towards achieving operational excellence in HSE objectives; ANSCPL shall:

- Adopt appropriate operational practices , safe system & suitable methods to monitor , control & minimize the risk & impact of the activities on HSE.
- Ensure that its operations at least comply with applicable HSE laws, regulations & statutory provisions .
- Ascertain that appropriate resources are made available to implement HSE policy & continuously review the policy 's relevance with respect to legal ,community & business development .
- Develop & maintain a motivated workforce trained for effective management of HSE emergency situations.
- Provide information on HSE policy to the concerned authorities & interested parties and conserve with business associates to facilitate their health and safety performance improvement with the intent of their individual HSE obligations.
- Cultivate a work culture where the HSE policy is understood, implemented & maintained by employees at levels within the organization.
- Continually improve its HSE performance by setting objectives & targets and conduct regular audits to ensure that operations are in compliance with HSE management system requirements.

Place : New Delhi

Date : 07/09/2017


HARDEEP SINGH

CHIEF EXECUTIVE OFFICER



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ALCOHOL AND DRUG POLICY

At ANS Constructions Pvt.Ltd. We are committed to excellence in all aspects of personal and professional life. We believe that Alcohol and Drug abuse is a significant obstruction to attaining this goal. Therefore, ANSCPL as policy does not permit the consumption of Alcohol of illegal drugs at the workplace.

Those violating this policy are subjected to disciplinary action. Drugs for medicinal purposes, however, can be administered by a medicinal practitioner.

Laws and traditions, as also religious secularity and political neutrality as a Company.

4. SAFETY DEFNATIONS

Safety is the state of being "safe" the condition of being protected against physical, social, spiritual, financial, political, emotional, occupational, psychological, educational, or other types or consequences of failure, damage, error, accidents, harm, or any other event that could be considered non-desirable. Safety can also be defined to be the control of recognized hazards to achieve an acceptable level of risk. This can take the form of being protected from the event or from exposure to something that causes health or economical losses. It can include protection of people or of possessions.

INCIDENT:

A chance event or condition that results in harm to people, damage to property or the environment

Loss of the industry.

NEAR MISS:

An unexpected or unsafe condition which not causing loss, injury or illness but can change in an accident.

FATAL:

Death resulting from an accident

FIRST AID CASES:

This is a non-reportable case, where the injured person is given medical treatment and



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discharged immediately for reporting on duty, without counting any lost time.

MAN-HOUR WORKED:

The total no. of employee-hours worked is called Man Hour worked. It includes managerial, supervisory, professional, technical, clerical and other workers including contractor, labors.

MEDICAL CASES:

Medical cases come under non-reportable cases, where owing to illness or other reason the employee was absent from work and seeks Medical treatment.

TYPE OF INCIDENT / ACCIDENT & THEIR REPORTING:

Non-Reportable Cases:

An accident, where the injured person is given medical helps and discharges for work without counting any lost time.

Reportable Cases:

In this case the injured person is disable for 48 hours or more and is not able to perform his duty.

Injury Cases:

These are covered under the heading of non-reportable cases. In these cases the accident caused injury to the person, but he still continues his duty.

HAZARD:

There are many definitions for hazard but the most common definition when talking about workplace health and safety is:

A **hazard** is any source of **potential** damage, harm or adverse health effects on something or someone.

RISK ASSESSMENT:

Risk assessment analysis process shall identify certain activities for which detailed written method statement or safe systems of work are required to ensure that the activities are properly controlled and executed safely, without risk to health and the environment. All such written safe



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systems of work shall clearly identify the objective, the sequence of operations, foreseeable hazards, and precautionary and protective measures required and shall be easily understood by the personnel who are to supervise and carry out the work.

FREQUENCY RATE:

Frequency rate is the no. of days lost for injuries per million employees-hours worked. Mathematically, the formula reads.

$$\text{FR} = \frac{\text{INJURY} \times \text{NUMBER OF LOST TIME}}{\text{TOTAL NUMBER OF MAN-HOUR WORKED}} \times 10,00,000$$

SAFETY AUDIT:

Safety audit of contractor's site will be carried out by HSE Department on Quarterly basis and report will be submitted to HSE DEPARTMENT.

HEALTH & SAFETY REVIEW:

A formal recorded management evaluation of the effectiveness of the Construction Health & Safety management systems and identification of actions for continuous improvement.

POSITIVE HEALTH AND SAFETY CULTURE:

The organization's attitude, values and beliefs with respect to the Health, Safety and Environment is a vital part of the process to achieve the aim of Zero Accident Performance. The creation of a positive culture towards the Health & Safety and Environment requires involvement and participation at all levels.

Good communication is required between all workers of organization.



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5. Safety objectives, Goal/targets.

SAFETY OBJECTIVE

- Provide a safe working environment.
- Ensure the safety of all personnel within our areas of work.
- Protect the public from injury.
- Prevent loss or damage to property resulting from our activities.
- Comply legislative and Company Safety rules in our areas of work.

Safety Goal

All level of ANSCPL Management, employees required to demonstrate a personal commitment to achieve excellence in project.

This will help us achieve our goals of

- Our goal is zero accident
- To operate incident and injury free.
- Ensure the environment is not adversely affected by our work
- To respect the community.
- To provide our clients with great experience.

Safety target

- Fatalities and lost time: Nil
- Near Miss : 05 per day
- Injurious/(LTD accidents) : Nil
- Fire: Nil
- Explosion: Nil
- Environmental incident: Nil
- Vehicle / transport incident: Nil
- Safety Audits: 2 Minimum
(Quarterly per year project whichever is high)
- Safety Training



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- Daily Site Safety Inspection
- Regular tool box meeting
- Job specific tool box meeting
- Weekly safety meeting
- Weekly site safety walk thru

SAFETY TRAINING

All Supervisor or worker will be imparted suitable safety training suitable

Safety training before they are deployed for work at construction site.

Record of the training programs along with list of participants will be kept.

Evaluation of safety awareness will be done after every safety program to judge

Effectiveness of safety programs & record will be kept.

A) TRAINING AND INDUCTION OF PERSONNEL.

All persons being inducted at site would be made to go through an Induction programmed of at least a one hour's duration. This should include.

- General rights and duties of workers at the workplace;
- Means of access and egress both during normal working and in an emergency;
- Measures for good housekeeping;
- Location and proper use of welfare amenities and first aid facilities
- Provided;
- Proper use and care of the items of personnel protective equipment;
- General measures for personal hygiene and health protection;
- Fire precautions to be taken;
- Action to be taken in case of an emergency;
- Requirements of relevant safety and health rules and regulations.

B) WORKERS TRAINING:

Each worker will be giving minimum 1hr. Initial safety training before

Deployment on job. Topics covered should include:

- HSE Policy
- General safety rules at site.



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- General safety precautions
- Accidents- Causes
- Fire Prevention
- Personal Protective Equipment.
- Safety during Construction.
- Action in case Emergency – Fire /Accident
- Procedure for Evacuation
- Evaluation by quiz, questionnaires

C) SUPERVISOR'S TRAINING:

Each supervisor will be given minimum two hour safety training pertaining to construction site.
Topic covered should include:

- HSE Policy
- General safety rules at site.
- Duties of supervisor in promoting Safety
- General safety precautions
- Accidents- Causes, Reporting & Investigation
- Work Permit system
- Fire Prevention
- Personal Protective Equipment
- Safety during Construction
- Electrical Safety
- Action in case Emergency –Fire / Accident
- Procedure for Evacuation
- Related chemical Hazards.
- Evaluation by quiz, questionnaires

D) REFRESHER TRAINING:

Refresher training will be given once in six months for every worker & supervisor,/Engineers



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DAILY SITE SAFETY INSPECTION:

Both the Work Site supervisors and Safety Supervisors are to conduct daily site Safety inspection around work activities and premises to ensure that work methods and the sites are maintained to an acceptable standard. The following are to form the common subjects of a daily Safety inspection:

- A Daily Safety Check Sheet if any Client shall also be filled in by Site Egg. & Counter Signed by Safety Officer.
- Personal Safety wears & gear compliance.
- Complying with site safety rules and permit-to-work (PTW).
- Positions and postures of workers.
- Use of tools and equipment etc. by the workers.
- Inspection should be carried out just when work starts beginning of the day, during peak activities period of the day and just before the day's work ends.

SAFETY INSPECTION:

- By RCM/DPM - Weekly
- Safety manager - Weekly
- By work Supervisor - Daily
- By Safety Officer - Daily

Tool Box Talks.

Site Engineers/Supervisors Toolbox talks would be conducted for the workers every day. These would cover:-

- a) Details of work for a particular day;
- b) Hazards involved;
- c) Precautions to be taken; and
- d) Safe procedure of the work to be followed.

JOB SPECIFIC TOOLBOX MEETING

- As with a job specific toolbox meeting, the subject matters must pertain specifically to a job being intended for immediate execution.



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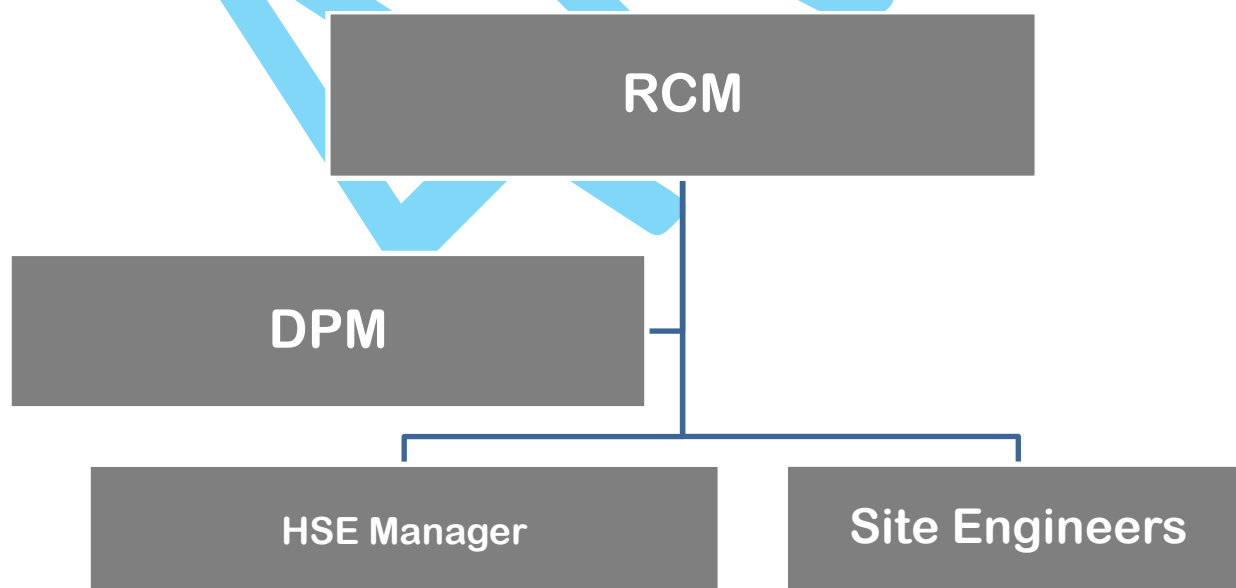
- The relevant Job Safety Analysis (JSA) should be drawn for a job specific toolbox meeting.
- The work supervisor shall be responsible to draw the relevant JSA or HIP for the meeting.

WEEKLY SAFETY MEETING:

- Regular OWNER/Contractor's Safety Inspection and meeting shall be an officially coordinated activity. Inspection findings and subsequent meeting M.O.M. shall be produced with copies extended to Owner if directed. The agenda to be discussed in the meeting should cover but not limited to the following:
- Good safety practices observed.
- Deficiencies, concerns found during all previous inspections and emphasize on what need and how to improve.
- Incidents/accidents or near misses that occurred.
- The inspection and meeting should be participated and attended by relevant supervisors nominated by Construction Manager on a rotation basis.

Individual responsibilities

6. Organization chart





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RCM:

The RCM will have overall responsibility of Safety, Health and Environmental at site for ensuring this, RCM is required to:- Be familiar with and demonstrate his commitment to implementation of The Company Policy and promote a continuous improvement in HSE Performance. Ensure that all personnel for whom they are responsible are aware of The Policy and implement relevant procedures as per requirement.

- Initiate formal hazard identification and risk analysis.
 - Ensure that a comprehensive site HSE Plan is maintained and complied with for the Project and distributed to all relevant parties.
 - Monitor HSE performance of the personnel and activities under his Control.
 - Establish a consultation and communication system with all employees and contractors'.
 - Ensure competence of subcontractors is verified in accordance with Safety standards lay down by the client. Ensure sufficient induction training for all employees and workers before commencement of work at site and subsequently for new Inductees.
- Comply with HMEI accident / incident investigation and Reporting procedures.

Ensure provision of adequate first aid Facilities and suitably trained Personnel.

DPM

The DPM will be responsible to the RCM for implementation of safety procedures. In the absence of RCM, he would take control of the Site. His duties are similar to that of the RCM.

Site Engineers

They will be responsible to the RCM /DPM for implementing the requirements of Safety procedures. In particular they are required to:-

- Be familiar with the Company Safety Policy, and client's health and Safety policy.
- Monitor all site operations to ensure that standard procedures are followed at site.
- Liaise and cooperate with site safety supervisor and ensure that defects Brought to attention are corrected.
- Establish and maintain proper communication with all workers with regard to safety aspects.
- Provide proper supervision for the work.



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HSE.

Accountable to the RCM for fulfilling the duties assigned to him in the safety policy and ensure implementations of safety plan. Safety Manager is also responsible to the RCM/DPM for Provision of health and safety advice and reporting on HSE performance of the Project. In addition, his duties will include:-

- Monitor and advise relevant personnel on compliance with HSE Statutory obligations at the site.
- Provide copies of site / office inspection and audit report to relevant Managers.
- Monitor compliance with accident, dangerous occurrence and Environmental incident reporting procedure detailed in the policy.
- Safety walks along with RCM weekly.
- Conduct accident, dangerous occurrence and environmental incident Investigation and produce reports detailing clauses, conclusions and Recommendations to prevent a recurrence.
- Undertake programmed of regular safety checks at site.
- Attend and participate in site safety committee meetings, management HSE review meeting and periodic safety advisors meetings.
 - Access and advice Site Managers and sub contract Managers on the Perceived HSE training needs of ANSCPL personnel.
 - Organize and conduct HSE training for site.
 - Submit monthly HSE report.
 - Monitor HSE performance of subcontractors and make appropriate Recommendations for performance improvement.

Employees.

All employees are accountable for conforming to the requirement of the Safety Plan and Policy and of statutory requirements. In particular every employee is required to:-

- . Take care of health and safety of themselves and others who may be affected by their acts or omission.
- . Co-operate with ANSCPL in fulfilling its statutory HSE obligations.
- . Co-operate with ANSCPL in pursuit of continuous HSE performance



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7. GENERAL SAFETY REQUIREMENTS:

- Where work activities may conflict with or impact upon other work activities take steps to discuss the matter with relevant parties and establish a consensus. Ensure that both parties and their teams are fully aware of potential hazards together with appropriate precautions.
- All plant and equipment shall be inspected before work commences and at periodic intervals thereafter. All plant and equipment shall be safe before and during use.
- Only competent personnel are permitted to operate plant and equipment and such persons shall be tested by a competent person and issued with an Appointment Certificate authorizing them to operate specified plant and equipment.
- Only qualified and appointed personnel (such as riggers and scaffolder) shall do skilled work. Such skilled workers shall always be identified by relevant band around hardhats (red band riggers and brown band for scaffolds).
- Method statements / risk assessments shall always examine the people, equipment materials and the environmental aspects of the proposed work. Such method statements risk assessments shall always be referred to sub-contractor / contractor safety department for review and comment.
- Ensure that an emergency / rescue plan to cater for any mishap (injury, fire etc.) is available, known to all people and capable of immediate activation. Maintain good housekeeping at all times. Provide and maintain adequate hygiene and welfare facilities such as toilets, hand washing measures drinking Water and canteen shelter, food waste bins and defined smoking area.
- A specific task related risk assessment shall be completed for this work activity and appended to the method statement. The risk assessment shall be complied on the approved format.
- Hazardous area should be barricaded with the attachment of appropriate warning symbols.
- Work area should be always kept clean. Unwanted scrap or tools should not be left unattended that may be hazardous to others.
- Personal protective equipment's should be used wherever required.
- All warning symbols should be well obeyed and regarded.
- It should be ensured that all-electrical cord: hoses and leads are well protected or elevated such that there are no obstructions to stairways



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8. WORK PERMIT PROCEDURE

WORK PERMIT SYSTEM:

Work Permit system as per client System Policy and client work permit system to be followed.

Permit to be maintain for 90 days and hot job permit copy should be submitted to HSE department daily. Permit audit to be carried out on weekly basis.

Work Permit Requirement Circumstances:

Work Permits are required for all work carried out within the Restricted Area as designated by the issue of standard form.

Work permit system are the most formal method of ensuring safe systems of working Their use shall be reserved for work where there are potential hazards and that the precautions necessary for that work need positive enforcement. Once work (radiography, pressure testing, confined spaces, instrument, fire gas alarm system etc.) has been identified and designated as requiring a permit to

There are two types of permits:

Cold Work Permit:

All work, which does not fall into the category of any other type of permit, must be authorized by a Cold Work Permit. Exceptions being that work which is done by certain Construction personnel in the course of their duties, all racking out or racking in or re-setting of equipment in electrical sub-stations by competent Electrical personnel only.

For exp. – Brick works, manually excavation, cable laying, material shifting, instrument work, loading and un-loading of material etc

Hot-Work Permit:

All work in the restricted area, which involves the use of any flame or electric arc or the use of any equipment likely to create a source of ignition must be authorized by a Hot Work Permit.



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Welding and burning equipment; electrical; tools: equipment with naked flames: soldering irons: sand or grit blasting equipment: pneumatic drills electrical and battery operated devices which have not been certified as safe for use in a flammable atmosphere: vehicles and other engine power mobile equipment.

Before the issue of a Hot Work Permit, the Issuing Authority must be in possession of an Atmospheric Test Certificate relating to the equipment or area on, or in which, the hot work is to take place.

Confined Space / Entry (Wherever required, permit/to be issued by client).

Blanket Work Permit:

It will not be issued in any Operational Area in any case.

Responsibilities for Permit:

Issuing Authority:

Is the person who signs a Work Permit and authorizes the work to start provided that all the prescribed special conditions have been or shall be complied with. He shall ensure that all supporting documentation has been obtained and is properly completed before the Work Permit is signed.

Performing Authority:

Is the person who receives' a Work Permit Normally a supervisor of the operatives carrying out the work. He shall ensure that he, and all the operatives involved, understand the conditions, limitations, and precautions necessary as stipulated in the Work Permit, and that these are complied with.

- The issuing Authority is responsible for ensuring that before a Work permit s issued.
- The work area a safe.
- Access to the work area is safe.
- The equipment involved has been made safe e.g. electrically isolated and locked; isolated from all process and utility systems; gas tested for hydrocarbons, toxic gas and oxygen contest, etc.
- Any additional form, e.g. 'Atmospheric Test Certificate' has been completed and attached to the Work Permit as required.



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Performing Authority:

The 'Performing Authority' is normally the Supervisor who is responsible for the work to be carried out and who made the application. He shall ensure that:

- He thoroughly understands the conditions and limitations of the Work Permit as issued and of the precautions necessary while carrying out the work.
- All personnel in carrying out the work are made fully aware of the conditions, limitations and precautions necessary as stipulated by the Work Permit.
- That the Work Permit and associated documentation are displayed or kept at the work's location.
- That the Work Permit and associated documentation are returned to the Issuing Authority at the expiry time or before then, if work is completed.
- That the Work Permit is renewed if work is required to continue beyond the time limit shown on the Permit.
- All work which involves digging or extraction of any surface / ground medium such as soil paved roads and concrete or the driving of stakes into the ground must be authorized by the issue of an Excavation Permit.
- The purpose of the Excavation Permit is primarily to protect buried services e.g. electric cables gas-water mains process lines etc. The Issuing Authority for an Excavation Permit shall have full knowledge of buried services etc. in the specific area.
- Electrical Isolation/Work Permit (Wherever required permit/to be issued by Client). Before any work is carried out on any electrical apparatus, it must be isolated and
- Locked, and an Electrical Isolation/Work Permit issued. In the case of an Electrical
- Isolation/Work Permit, the Permit must be countersigned by a competent Electrical Engineer. The Electrical Isolation/Work Permit is for work on 440V or below apparatus and associated equipment. For work involving higher voltages, special arrangements shall have to be made with the Local Electrical Distribution Authority and these arrangements shall be outside this permit system. For work on associated equipment e.g. a motor driven pump, agitator, cooling fan etc., a motor isolation tag must be issued and posted at the driver. Personal padlocks must be issued to all directly involved operatives working on the associated equipment for them to insert in the locking device on the electrical isolator.



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Acknowledgement:

All Work Permit categories require a formal acknowledgment from the Recipient where he is required to sign the Permit Form acknowledge that he understands all the conditions precautions and restraints imposed by the Issuing Authority and that these conditions etc., have been fully explained to the operatives who shall carry out the work. The 'Acknowledgement section shall be signed at the time of issue.

Posting, Suspension, Completion and Cancellation:

While the work is in progress the Permit together with any supporting documentation-

When all work is finished, the person in charge of the work normally the recipient of the Work Permit shall sign the 'Work Completed' section of the Form and return it along with any supporting documentation, to the Issuing Authority. When the Issuing Authority has received the completed Permit it shall automatically be cancelled.

Validity:

A Work Permit is only valid until the time/date shown. Normally a Work Permit shall be issued for a period not exceeding 08 hours. Each Permit may be extended on a 08-hour basis. The permits shall be presented for re-validation each day before work commences. A Work Permit ceases to be valid upon the sounding of an emergency alarm.

9. PERSONAL PROTECTIVE / SAFETY EQUIPMENT:

As a minimum dress code all Personnel shall be required to wear hardhat safety footwear, long trousers, short sleeved shirts (or overalls) on construction sites. This shall be emphasized during the site induction training.

Personnel not in possession of the appropriate work shall not be allowed to work on site. Hard hats shall display details of the individual (Name / Occupation) and the company employing him and shall be available before an individual's induction training. Where a sub-contractor does not have a corporate logo then the logo of the main sub-contractor ANSCPL shall be used. In certain work activities, it may be a mandatory requirement for overalls



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(coveralls) to be worn. Where there is need for additional personal protective equipment, safety spectacles, hearing protection etc. this shall be identified in hazard-risk assessments. PPE to be issued to the individual workers

RELEVANT IS-CODES FOR PERSONAL PROTECTION

IS: 2925 – 1984	Industrial Safety Helmets.
IS: 4770 – 1968	Rubber gloves for electrical purposes.
IS: 6994 – 1973 (Part-I) Gloves). IS: 1989 – 1986 (Part-I-II)	Industrial Safety Gloves (Leather & Cotton Leather safety boots and shoes.
IS: 3738 – 1975	Rubber knee boots.
IS: 5557 – 1969	Industrial and Safety rubber knee boots.
IS: 6519 – 1971	Code of practice for selection, care and repair of Safety footwear.
IS: 11226 – 1 IS: 5983 – 1978	Eye protectors.
IS: 9167 – 1979	Ear protectors.
IS: 3521 – 1983	Industrial Safety belts and harness

GENERAL REQUIREMENTS

The risks posed in any particular work activity shall be assessed, and adequate PPE selected in accordance with the following:

- Gives protection against risk(s) without in itself leading to any increased risk
- Suitability for the user
- Compatibility to the work activity

All employees shall be provided with the necessary PPE, for their particular work activity. PPE and the necessary training courses shall be supplied to the employee.

All employees shall be held responsible for the proper care and use of any PPE supplied to them.

Supervisors shall be responsible for ensuring that all personnel on the project are trained in the use of, provided with, and wearing all PPE required for the activity and environment of the work. Personnel not properly equipped, for whatever reason shall not start or continue work



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Selection of PPE

PPE shall be selected and used to provide protection for all personnel against

- Inhalation and respiratory tract hazards
- Skin contact hazards
- Mechanical injury and hazards
- Construction safety hazards
- Physical agent hazards
- Environmental hazards
- Radiological hazards

Consideration of the hazards introduced by PPE ensemble selected and the consequences of the use of an ensemble shall be included in the determination of the appropriate for each work element

10. FIRE PREVENTION AND CONTROL

1. Water and sand buckets shall be kept full near all hazardous locations.
2. Fire fighting training to be conducted for all, specifically supervisors/foremen.
3. Suitable number of extinguishing equipments shall be kept at strategic locations.
4. Extinguisher of oil fire and electric fires shall never be attempted with water. The same to be done with foam cylinders or with sand and earth.

All necessary measures should be taken to:

- . Avoid the risk of fire;
- . Control quickly and efficiently any outbreak of fire;
- . Bring out a quick and safe evacuation of persons.
- . Inform unit/fire station control room, where construction work is carried out within existing operating area.

Basically fire is a chemical reaction. The following are necessary for fire to take place:



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1. Fuel 2. Heat 3. Oxygen

To eliminate the causes of fire, it is important to locate how and where fire starts.

Classification of Fires:

- CLASS A FIRES Wood, Textiles, and paper rubbish and the like;
- CLASS B FIRES Oil, Petroleum, Solvents, Grease Paint and the like;
- CLASS C FIRES Gaseous substance under pressure;
- CLASS D FIRES Reactive chemicals, active metals and the like;
- CLASS E FIRES Electrical equipment, delicate machines and the like.

Fire Extinguishers:

The exact number and type of fire extinguisher for each site will be decided at Start of work by site in-charge depending upon the requirement. All fire Extinguishers will be kept in good working.

11.SAFETY PROVISIONS FOR PLANT, MACHINERY, EQUIPMENT AND HAND TOOLS

Safe operating procedures should be established and used for all plant, machinery and equipment.

HAND TOOLS:

Accidents arising out of hand tools can be attributed to any one of the following reasons:

- A - Using the wrong tools;
- B - Using tools, which are in poor condition?
- C - Using the tool in a wrong way;
- D - Keeping tools in unsafe places.



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If the above four conditions are taken care of we can eliminate all the hand tool accidents.

JACKS:

- Select jacks heavy enough to raise and hold the load safely.
- Jacks of same capacity and type should be used while using number of

Jacks

- Load must rest on firm packing before releasing the jack or before allowing Persons to work below the raised load.
- Inspect frequently and use only the proper grade and clean oil. It is Advisable shore up any load that must remain in a raised position for any Length time.

PORTABLE ELECTRIC TOOLS:

1. Maintenance of electric tools should be systematic.
2. Safety guards provided in the tools should not be tampered with.
3. Gloves, safety shoes, goggles, etc. Should be worn by the operator wherever Necessary.
4. Only experienced and authorized personnel should be permitted to operate Power tools.
5. For all electric power tools, a running earth must be maintained and the Supply cable should be handled very carefully.
6. All cabling for Portable tools of switches / ploughs should be kept in proper Condition. These should be checked for safety once in 3 months.
7. Electric supply should be disconnected before attempting any repairs or Servicing. Even a change of wheel in the grinding machine requires the Supply to be disconnected.

DRILLING MACHINE:

1. A prick punch or pilot hole should always be provided to guide the drill bit.
2. Suitable drill bit should be selected for the material being drilled.
3. If bit is long enough to pass through the object, care should be taken to avoid



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Damage or injury on the other side.

4. If the object is small, it should be secured to prevent spinning.
5. Care should be taken to prevent sleeves and other clothing from being wound Around drill.

PORTABLE GRINDING MACHINE:

1. Hood guard provided in the machine should be maintained in place always.
2. Wheels of proper 'rpm' rating should be used.
3. The grinding wheel shall be stored and handled properly.
4. Only experienced and skilled grinders shall be engaged.
5. The grinding machine shall not be allowed on the ground when the wheel is rotating.

BENCH GRINDERS:

1. It should be provided with a proper earthing.
2. Eye shield and hood guard should be provided and maintained.
3. Wheels of proper 'rpm' should be used and they should be carefully inspected to check for cracks.
4. Tool rest should be in place properly adjusted to a maximum gap of 1/8 Inch. From the wheel.

PNEUMATIC TOOLS:

1. Air hoses of Pneumatic tools should be protected against whipping.
2. The airline should be de-pressurized before opening any joint.
3. Compressed air should not be directed against self or others.
4. Air hoses taken overhead or vertically should be sufficiently supported.



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PULLEYS:

- 1 Proper pulley should be used, according to the requirement of work.
- 2 In no case pulley meant for Nylon rope should be used with steel rope.
3. Heaves, shaft, hook, hook pin, lock of pins should be checked before use and lubrication did on necessary parts.
4. Grooves of the Sheaves should be uniform and smooth. The wire rope or fiber Rope should run free without touching against the block or suspension parts.
5. Heaves should be strong and firm.
6. The shaft should be free from crack and should not be worn out.
7. Anchorage should be strong and firm.
8. Anti-water should be used to prevent rubbing of ropes against one another.
9. Workmen should remain clear of all loads, when lowered by pulley / cranes etc.

CHAIN BLOCK / PULL LIFT MACHINE:

1. Chain blocks of proper lifting capacity supported by Test Certificate should be used for lifting known loads.
2. Chain block must be checked, and tested periodically. It should be lubricated before every use.
- 3 .Do not cannibalize parts from a chain block to another.
4. Chain blocks should be tested for slip by suspending safe load.
5. It should operate freely and the chain should not come out of pulleys
6. The anchorage should be strong and rigid.
7. They should be checked for cracks, excessive wearing, elongation, etc. Hooks opened out should not be used.
8. No chain block / pulley which has been tampered, be used unless it is thoroughly checked and tested by competent person



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9. Chain block / pulley must be checked if stored for longer time, by subjecting To shock load, to observe slipping of load / jamming of link.

WINCHES:

1. Safe working load with gearing arrangement should be marked on the winch

And tested.

2. Winch should not be over-loaded.

3. It should be placed on a firm base and properly anchored.

4. The brake, ratchet arrangement, gear and pinion including the meshing, wire

Rope and its clamping arrangements and direction of receiving rope drum; tie Rods should be checked before using the winch.

5. Ratchet arrangement should be kept in position while hoisting a load.

6 Tie rod should be adjusted not to allow drum movement causing clutch Arrangement to slip.

DERRICKS:

1. The mast, guy-ropes, wire ropes, swivel hook, rope clamps etc. of the derrick

Should be ascertained by a competent person and it should not be over
Loaded.

2. The mast, guy-ropes, wire ropes, swivel hook, rope clamps etc, of the derrick Should be thoroughly checked before erecting the derrick.

3. Depending on the type of derrick, the mast should be firmly fixed by guy ropes / structure.

4. All precautions should be taken so that base of the derrick does not shift or Sink.

5. Guy-ropes of the derrick should be anchored tightly with strong structures /Hold fasts / anchorage blocks.

6. The load being hoisted should not run against the derrick.

7. All welded parts of Derricks (especially in bracing and stiffener) should be

Periodically checked for any crack, or defects in metal it.



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8. If bolted joints are used, check for proper bolts and their tightness.

MIXERS / VIBRATORS:

A. **Concrete Mixers:** All gears, chains and rollers of concrete mixer should be adequately guarded to prevent damage.

2. Concrete Mixer hopper shall be protected by side railing to prevent workers from passing under them and operators shall make sure before lowering the skip that all workers are in the clear.

3. Hopper hoisting wire rope has to be checked for its condition periodically.

4. Hopper hoist and anchoring brake should be checked / adjusted.

5. Skip hoist and anchoring brake should be checked / adjusted.

6. Nothing should be kept inside the motor enclosure.

7. be sure that Motor fan guard is secured firmly.

8. be sure that wiring is properly connected and insulated.

9. Ensure double earthing is done for Electric mixers.

B. Concrete Vibrators

1. Vibrating unit shall be completely enclosed and belt transmitting the power to the unit adequately guarded.

2. Electrically operated Vibrators shall be totally enclosed and be protected against overloads by suitable overload relays and shall be effectively earthed.

3. be sure that sufficient length of cable is provided to the Vibrator.

4. Ensure electric starters are fixed firmly on the stand.

5. While needle is inserted in the vibrator, be sure needle load is firmly locked.

6. be sure to lubricate needle inner core.

7. Vibrator operators should wear Rubber soled shoes.



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PNEUMATIC COMPRESSOR:

1. Compressor should be checked for any crack in the Air receiver and pipe Connection.
2. Safety valves should be checked periodically for their effective functioning at set pressure.
3. Pressure gauges to be checked.
4. Compressor should not be installed / placed near excavation pits / blasting Area.
5. Air hose ends should not be kept free to avoid whipping.
6. No one should use compressed air to clean dress or body. It should not be directed on others. No horseplay is allowed with compressed air.
7. Supply of compressed air to pneumatic tools should be controlled by valves and connected with oil lubricator.
8. Goggles should be used.

C R A N E S :

1. The capacity of the crane should be ascertained before using. Brakes should be checked while lifting critical load and adjusted if needed.
2. Crane should never be overloaded.
3. Mobile cranes should be parked on hard soil or strong base. They should not be placed near the edge of the pit or excavation.
4. Safe working load of any mobile crane depends on;
 2. . Condition of the ground
 3. . Boom length
 4. . Radius of rotation while lifting the load inclination of boom to the vertical.
 5. . Out rigger blocked / free
6. The safe working load is generally tabulated in the Load Chart of the crane. Sometimes cranes are de-rated due to the defects in welding, bend in angle, bracing etc. and condition of clutch, brake etc.
7. The load is the total load hung from the rope Heaves of the boom including weight of hook block, ropes / slings, etc.
5. Standard code properly understood by the operator and trained signalman
8. Should be used. The Crane operator shall respond to signals only from the appointed signalman but shall obey stop signal at any time no matter who gives it.



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6. Tag lines should be used while hoisting heavy and bulky materials.
7. The brakes, boom, hook, wire ropes and pulleys, rope anchoring should be checked periodically by a Maintenance person to ensure the Crane's safe operation.
8. The load being lifted should not touch the boom.
9. The boom or any part of the crane should not come near any live electric line / service line.
10. Swinging of load should be done smoothly.
11. Proper quality of packing should be used and the outrigger should rest tightly on the packing.
12. Nobody should stand below the boom or load.
13. The operator should be able to see the hook and load throughout the hoisting period.
14. During storm the hook block should be anchored firmly and swing lock be released.
15. When an extended boom is used on the crane, the operator must use extreme care in lowering the load to the ground. An extended boom never should be lowered to one side of the chassis as the stability of the crane is usually reduced in that position and the crane will get upset.
16. The crane has to be moved on a heavy timber mat whenever there is instability of soil.
17. The use of any makes shift methods to increase the capacity of crane such as Timbers, with blocking, or adding counter-weight, should not be permitted.
18. Before leaving the crane at the end of the workday, the crane operator should Remove the load from the hook and raise the hook block to maximum height.
19. The crane operator should keep the deck clean of any oil, mud and grease.
20. Operator should always keep the windshield clear, in order to prevent accident due to this.
21. Ensure at least two full winding of ropes be always on the rope drum. After a
22. Boom extension, the hooks shall be lowered to the required lowest point to
23. Ensure that at least two dead coils remain on the drum and to the highest
24. Point to check that the drum capacity will not be exceeded.

POWER SHOVELS (EXCAVATORS) :

1. Power shovels (excavators) shall be so operated as not to lose their stability.
2. Power shovels that are equipped with unit for deep digging shall either be so designed that the bucket teeth should not come nearer the undercarriage than 40 cm. or be provided with reliable stop that prevents the bucket to come nearer than the specified distance.
 - 2.1. The boom of the power shovel (excavator) shall be prevented from accidentally swinging during operation transport.
 - 2.2. The boom shall not be pulled tight against the emergency stops while supporting a load.



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2.3 The maximum boom length stipulated by the manufacturer shall not be exceeded. The wire ropes used for the boom suspension as well as for the shovel / bucket shall be of the specified diameter and construction.

3. The safe operating radius of the load under consideration shall not be exceeded.

4. While operating power shovels the driver shall ensure that no person is working, passing or standing under the raised bucket or grab.

5. The bucket or grab of a power shovel shall be pulled out of the bank as soon as it is full. When not in use, the bucket shall be kept resting on stable ground and shall not be left hanging.

6. The bucket or grab of a power shovel shall be fixed to restrict movement while it is being repaired.

7. While operating near edge of excavations or embankment substantial space shall be provided to prevent it from approaching a dangerous position and the sides of the excavation shall be adequately shored. Heavy equipment such as excavating machinery and road traffic shall be kept back from the excavated sides at a safe distance.

8. Power shovels shall not be left on a slope with the Engine or motor running.

9. The height of benches in overburden shall not be more than the height of the boom of the machine used for digging excavation or removal.

12. JOB SAFETY PROCEDURES DURING VARIOUS OPERATIONS

EXCAVATION

1. Before starting any excavation of any description in the existing plant / Factory area premises, Work permit must be obtained from the Clients to avoid any damage to the underground electric cables or pipelines.
2. Safe angle of repose while excavating trenches exceeding 1.5m depth up to 3.0m should be maintained. Based on site conditions, provide slope, usually 45 degree and suitable bench of 0.5m width at every 1.5m depth of excavation in all soils except hard rock or provide proper shoring and Strutting to prevent cave-in or slides.
3. As far as possible, excavated earth should not be placed within one meter of the edge of the trench or depths of trench whichever is greater.
4. Don't allow vehicles to operate too close to excavated area. Maintain at least 2m distance from edge of excavation. No load, plant or equipment should be Placed or moved near the edge of any excavation where it is likely to cause its Collapse and



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thereby endanger any person unless precautions such as the Provisions of shoring or piling are taken to prevent the sides from collapsing.

5. 5. Adequately anchored stop blocks and barriers should be provided to prevent Vehicles being driven into the excavation. Heavy vehicles should not be allowed near the excavation unless the support work has been specially designed to permit it
6. If an excavation is likely to affect the security of a structure on which persons are working, Precautions should be taken to protect the structure from collapse.
7. 7. Barricade at 1m height (with red & white band /self glowing caution board) should be provided for excavations beyond 1.5m depth. Provide two entries/exits for such excavation.
8. 8. Necessary precautions should be taken for underground utility lines like cables; sewers etc. and necessary approvals/clearances from the concerned authorities shall be obtained before commencement of the excavation job.
9. Dewatering shall be done if any accumulation of water in the trench. Necessary precaution should be taken to prevent entry of surface water in trenches.
10. During rains, the soil becomes loose. Additional precaution shall be taken to prevent the collapse of side wall.
11. In case of blasting, follow strictly is: 4081-1986 & Indian explosive Act and rules for storage, handling and carrying of explosive materials and execution of blasting operation.
12. Pick and shovel people working in excavations should be kept far enough apart to prevent injury to one another.
13. All workmen working inside the pit shall necessarily wear safety helmets & safety shoes

SAFETY MEASURES FOR EXCAVATION

- a. Trenches 1.5 meters or more depth shall be sloped back to a safe angle of 45degree. Excavation in unstable soil may require shoring or additional sloping.
- b. All excavated materials shall be placed at least 1 meter from the edge of excavation.
- c. Safe access must be provided into all excavation by means of ladders. Stairs or ramps.
- d. Excavation beyond 3 meters shall have two entries/exits.
- e. All blasting operations shall be done by an experienced and authorized person only.
- f).Indian explosive act and explosive rules of the state and central rules and regulations shall be strictly followed for storage, handling, carrying explosive materials to site.
- f. Necessary precautions like barricading with red flags, sirens blown few minutes before start of blasting shall be followed.
- g. Excavation beyond 1.5 meters depth shall be barricaded and proper lighting



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arrangement shall be made.

- h. Necessary precautions at the edge of excavation to be taken where machines are in operation and also to protect entry of water in trenches.

BACKFILL

- Backfill operations shall be carried out in accordance with the following guidelines:
- Backstops will be provided for vehicles dumping material to prevent them over running the edge.
- Consideration will be given to the specification for compaction qualities, need for tile markers and tiles and other protection measures necessary for the protection of equipment.
- Redundant spoil not back filled will be removed to an area identified in advance.
- A safe system of work will be identified and implemented for the removal of timber supports, steel Sheet piling and piles. Consideration will include provision of lifting appliances suitable to take the load of embedded Sheet piling without damage.

WORKING AT HEIGHT

INSTRUCTIONS:

The primary means of achieving safe working conditions when working at heights is to provide adequate and sufficient access and egress arrangements and suitable working platforms at the place of work.

A Tool Box Talk will be given to each person carrying out work at height. Following an assessment of the risks, supervisors will instruct personnel on the procedures and precautions to be followed when working at height.

Approved full body safety harnesses should only be used as a last resort

Where conditions make it impracticable to provide a safe working platform.

Fall Protection Equipment

All personnel working above ground level shall be provided with, and shall use, appropriate fall protection equipment and PPE. Appropriate regulatory standards must be observed when using fall arrest equipment. Whenever falls could result in impalement or other injury e.g. working over chemicals, operating machinery etc. fall arrest equipment must be used.

- A competent person must regularly inspect fall protection equipment, at least every six months. It should also be maintained so that it remains satisfactory for use during the



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construction period. Effective actions must be taken to rectify any defects observed as a result of these inspections.

- A competent person and repaired or destroyed, as necessary must examine fall protection equipment, which has been deployed in a fall.
- Fall protection devices and systems shall not be used for any other purpose than those for which they are designed.

Safety Harness/Lanyard Systems

Fall protection in the form of safety harnesses and lanyards must be used in situations where it is impracticable to provide primary systems.

- Whenever full body safety harnesses are used they must be secured to a secure anchorage point, running line or arrester device.
- Full safety harnesses with thigh and shoulder straps shall be worn. The lanyard

And anchorage point should limit the maximum drop to 2 meters.

- Shock absorbers must be provided with the harness system in order to reduce the shock loading in the event of a fall.

Running Lines:

Running line systems can be used to provide a secure anchorage point for safety harnesses.

- Running lines should only be installed and maintained by competent persons.
- These systems must be designed for loads, which it is expected to take safely. The line must be (i) supported on anchorage's that are sufficiently strong (ii) securely fixed to these anchorage and (iii) of a sufficient size and strength to support a falling person or persons.

Self Reeling Arrestor Devices:

- Inertia reel devices, which can be directly attached to safety harnesses and anchor points, provide good protection against falls.
- These devices shall be securely anchored using shackles or karabiners attached to wire-rope slings or webbing slings.
- Regular inspections and checks of these devices shall be carried out in order to ensure



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that they operate efficiently.

- These devices provide fall protection to personnel engaged in the installation of other fall protection systems e.g. running lines, scaffolding etc.

Other Arrester Devices:

There are other arrester devices which are not self-reeling available for use where anchorage points are limited. Such systems may allow free vertical movement as well as movement around corners and past obstacles without the need to unhook safety harnesses.

SCAFFOLDING

A. Steel Scaffolding:

1. Before starting the scaffolding erection, the surface on which it has to be erected must be made firm and level.
2. Joint pin connects the pipes internally. This can be used only to connect vertical pipes.
3. Sleeve couplers are fixed on the outside of two tubes to connect them. This can be used to connect vertical, horizontal and slanting pipes.
4. DO NOT USE PIPES WHICH are -
 - (a) Split along their length
 - (b) Bent or kinked,
 - (c) Thin at the ends,
 - (d) Heavily pitted and
 - (e) Having badly cut or ragged ends.
5. Do not use corroded or deformed couplers and make sure that the threads are in good condition.
6. Dropping or throwing materials from the top and from bottom to top should not be entertained. A rope shall be used for that purpose.

B. Common Faults in Scaffolding:

- Supporting of boards inadequate and therefore liable to tilt.
- Absence of toe-boards and guard rails where necessary.



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- Faulty alterations made without approval.
 - Erected on uneven ground.
 - Supported by scaffold lashings instead of wire ropes.
 - * Couplers misused; use of ordinary coupler where load bearing coupler to be used.
 - Absence of ties where necessary.
 - Foundations insecure.
 - False supports, for examples, drums, ladders, piles of bricks, etc.
 - Outdated and damaged couplers.
 - Lack of bracing.
 - Defective boards, large knots, splits, etc.
 - Not wide enough - three planks is normally the minimum.
 - Gangways obstructed.
1. All scaffoldings shall be visually inspected for their stability, anchoring with permanent structures.
 2. All load bearing scaffoldings shall be approved (for the maximum likely load) and inspected during erection.
 3. Scaffolding planks shall not be less than 5 X 250 mm full thickness lumber or equivalent. These shall be cleared or secured and must extend over the end supports by at least 150 mm and not more than 300mm. No loose planks shall be allowed.
1. Suspended scaffolds shall not be less than 500 mm wide tied properly with ropes.
 2. Check the proper tagging is provided at the foot of scaffolding as.

Green Colour Red Colour

FORMWORK

1. Tubular steel frames used as staging to support concrete formwork should have a safety factor of at least 2 & be used in accordance with each manufacturer's recommendation.
2. Before erection of steel frame staging is started, a thorough inspection should be undertaken on it.
3. Struts and or diagonal braces must be in proper position and secured for frames to develop full load carrying capacity.
4. As erection progresses, all connecting devices should be in place and fastened for full stability of joints and units.
5. As additional heights above two tiers are added, suitable planking should be used as a working platform, which should consist of three 1.5 inches x 6 inch planks



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- minimum.
6. The capacity of the soil for foundations should be determined for every staging job. The effect of weather conditions should also be taken into consideration as dry clay may become very plastic after a rainfall and show a marked decrease in load carrying capacity. Care should be taken not to disturb soil used for foundation supports.
 7. Timber jacks, joists, stringers and ledgers should be inspected for defects such as cracks / excessive knots.
 8. During concrete pouring operation, there should be constant inspection of the staging system with provision for corrections as necessary.
 9. Before the reinforcement and shuttering works are started for Beams, Working platform / walkways are to be provided.
 10. Before starting the reinforcement work on concrete column, the existing scaffolding must be extended to more than the required height, so that it provides a means to anchor safety belts used by the workmen.

DUMPING, PILING AND STACKING

1. Dumping, piling or stacking of materials should not block roads, Passageways/walkway, escape route.
2. Material should not be dumped against existing walls or partitions to a height that may endanger the stability of the walls.
3. While withdrawing piled materials like loose earth, crushed stone, sand, etc from the tockpiles, no overhanging is allowed.
4. No material shall be so stacked or placed as to cause danger or inconvenience to any person or public or any other agency at work.
5. Dumping, piling or stacking of materials below overhead transmission lines should be voided.
6. No material should be piled, dumped or stacked at random but only in areas Specified for the same.
7. While removing materials from stacks, the sequence of removal should be only from top.

CONFINED SPACES

- Confined spaces could be in vessels, tanks, reactors, pipelines etc.
- Gas cylinders shall not be allowed inside confined spaces.
- A stand by man shall always be standing outside whenever any person goes for a work inside a confined space. A suitable communication system shall be adopted.
- Adequate ventilation shall be arranged by using exhaust fan and / or oxygen level inside



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shall be monitored.

- Only low voltage (24V) lamps shall be used for lighting.
- Tools with air motors or electric tools with maximum voltage of 24V shall be used.
- Make sure that the connected piping of the equipment which is to be opened is pressure free, water has been drained and vents are open.

FABRICATION

1. A space should be earmarked for fabrication yard.
2. All equipment electrically operated must have proper earthing and guards in position.
3. The use of rubber gloves or leather gloves in dry condition is an absolute must while operating hand held electric tools.
4. Drilling machine, grinding machine should have safety guards in position. If for any maintenance purpose guards are removed, it should be put back immediately after repair.
5. While fabricating heavy structures, adequate props should be given to avoid toppling of the component under fabrication.
6. Material should not be dumped haphazardly. It should be stacked properly in order.
7. Wear proper protective clothing and equipment such as goggles, leather gloves, face shield, etc. as protection against radiating heat and spark.
8. Do not watch Arc welding operation with naked eyes.
9. When welding or gas cutting is performed in confined space, the cylinders should be kept outside.

WELDING

Major risks involved in welding are electrical shock, burns, fire hazard, light radiation and fumes.

A. Avoiding electric shock:

Electric shock may arise due to poor earthing. Though the operating voltage during welding is less, any voltage more than 24 V is not considered as safe. The shock hazard mainly depends on the body condition i.e. when the welder is soaked condition due to perspiration or when he standing on wet places, or when his gloves, shoes are wet, the severity of shock is likely to be more.



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There will be one work lead and a return lead. These two leads should be maintained without any break in between with all rigid hoists, if any. In addition to this, the job should be grounded to the main earth and also a body earthing for the welding machine is to be ensured. Earthing for the welding machine is to be ensured. Earthing from job should never be connected to charged pipelines and running plant equipment.

The welding machine and the cables should not be kept in wet places. The inspection lamp provided to the welders should be preferably of 24 V type to reduce the shock hazard. Gloves should not be taken as substitute for holder insulation.

B. Avoiding Fires:

All combustible materials such as oil, paint rags etc. should be cleared off where welding operations are likely to be taken up.

While doing hot work (welding, cutting etc) in operational process plants, proper care should be taken.

While suspending the operations, even temporarily, the holder should be hung safely to avoid inadvertent contact with the structure and cause spark. The supply should be disconnected when the welder wants to leave the place of work even for a short while.

At the end of the job, before leaving the place of work, smoldering fires, if any, should be put out.

The welding transformer body temperature should not be allowed to exceed 85 degree C. Otherwise it will lead to fire hazard. The transformers should be kept in well-ventilated area to facilitate cooling.

All cable joints should be very rigidly made and properly insulated to avoid the cables getting heated up or producing sparks causing fires.

C. Avoiding burns. Heat effects:

There will be radiant heat while working on preheated job. To reduce the effect of radiant heat, asbestos curtain / apron should be used.

There should not be any pocket in welder's dress and the shirt sleeves or the pant cuff shall not be folded back giving a chance to the sparks to get retained here burning the dress and the skin.

D. Avoid radiation and fumes:



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Eye injuries are most common in welding. It could be caused:

- Due to seeing the arc welding rays directly / indirectly;
- Due to seeing the gas welding and cutting flames with naked eye;
- Due to fumes that emanate during welding;
- Due to fall of flying materials while removing excess metal and slag.

A welding shield or welding helmet made of fiber-glass, dark in color and fitted with shatter proof lenses of not less than 3 mm thick. If the surface of the metal to be welded is galvanized, the fumes evolved during welding can cause a temporary sickness.

Also metals such as lead, chromium etc. can give off very toxic fumes. In such cases, exhaust / ventilation must be adequately provided.

The main hazards likely to be encountered during welding and burning processes can be identified as follows:

- . High temperatures causing risk of fire and burns.
- . Fumes emitted from the process likely to cause injury to health.
- . Radiation causing 'arc eye' and burns.
- . Oxygen deficiency or enrichment, particularly within an enclosed space
- . Electric shock
- . Explosions during the welding and cutting of drums, tanks or vessels, which
Have contained flammable materials.

LIFTING

INSTRUCTIONS

Accidents involving the failure and damage of lifting equipment are frequent on construction sites and usually very severe when they do occur.

Causation factors leading to such accidents include:

- . Operator error



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- . Inadequate inspections and maintenance of equipment
- . Failure to plan the operation and take precautions against overhead power lines, poor ground conditions and other obstructions.
- . Overloading and misuse of equipment.

Lifting Operations Check List

- .Has the persons responsible been adequately trained?
- .Has the weight, center of gravity and nature of the article has been Determined?
- . Where is the article to be lifted from?
- . Where is the article to be moved to?
- . Has the crane site been determined and is it in the best location?
- . Taking into account the length of slings etc. and the clearance between the load and Crane jib.
- . What is the maximum load that can be lifted?
- . What is the maximum radius that the crane can operate when lifting the load?
- . What is the maximum (including slings/spreader beams etc.) that the crane will be lifting?
- .Is there enough space to erect and operate the crane and maintain a safe distance between the counterweight and any fencing, building etc? To enable pedestrian access.
- . Are there any slew limitations on the machine, when under load?
- .Has the route been checked for sitting the crane with regard to any overhead obstructions?
- .Are there any overhead power lines or other elevated obstructions, which the crane is likely to come into, contacted with.
- .Has the driver and the crew involved in the lifting operation been fully briefed.

13. COMPRESSED GAS CYLINDERS

A. Unloading of cylinders: (No Gas cylinders will be accept without cap)

1. From the truck, unload directly on a raised platform by rolling over a coir mattress. If a



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suitable raised platform is not available slide down each cylinder over a 15 cm thick reinforced coir mattress, taking care that the bottom end touches first.

2. Lifting magnet should not be used for loading or unloading.
3. Cylinders should not be loaded loosely in a vehicle, failing which it will be subjected to heavy jointing and damage during the vehicle movement.

B. Storage:

1. The cylinder storeroom must be well ventilated.
2. Full and empty cylinders are to be stored separately.
3. Acetylene cylinders should be stored upright and properly secured.
4. Other cylinders, if stacked horizontally, proper chocks should be used to prevent rolling.
5. Do not keep cylinders in battery room or oil storage room.
6. Storage cylinders well away from sources of heat.
7. Oxygen and Acetylene cylinders should be stored separately.

C. Care in usage of cylinders:

1. Oil or grease should not come in contact with the cylinder valve assembly or the regulator fittings.
2. Do not use cylinders as rollers, work supports or jacks.
3. Never allow cylinders to come in contact with electrical apparatus or live wires.
4. Do not use chain slings for lifting cylinders; only fiber sling should be used.
5. Use red hose for acetylene and other fuel gases and black / green hose for oxygen. Never use hoses of equal length and do not coil the hoses around regulator or cylinder.
6. Nozzle tips shall be kept clean always; otherwise backfire may result.
7. Set the regulators to the recommended pressure and ensure leak-free connections.

‘COMMANDMENTS’ Regarding Gas Cylinders

1. Do not issue a cylinder to site unless contents are clearly identified.



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- .Do not use a gas cylinder unless contents are identified.
- .Do not handle cylinders or valve assemblies with greasy hands or oily rags.
- .Do not lubricate cylinder valve threads.
- .Do not lubricate cylinder valve threads.
- .Do not stack cylinders near sources of heat or indirect sun.
- .Do not lay cylinders direct on wet soil.

CONCRETING

- a) Safety cages shall be provided around moving motors etc.
- b) Vibrator hoses shall be in proper working conditions and mechanically locked.
- c) Workers working with cement shall have wear respirators.
- d) Electrical mixers, vibrators etc shall be suitably earthed.

REINFORCEMENT

- a) Workers to use hand gloves, safety shoes helmets etc.
- b) For supplying or fixing of rods at heights, proper staging, bundling to be provided.
- c) For supplying distance carrying of materials on shoulders, suitable pads to be provided to workers.
- d) While transporting by trucks/trailers, the rods should not protrude in front of or by the side of driver's cabin. In case such protrusion cannot be avoided behind the deck, then it should not extend 1/3rd of deck length or 1.5 M whichever is less and tied with red flags/lights.

14. Labour Welfare

Drinking Water:

In every workplace, there shall be provided and maintained at suitable places, easily accessible to labour, a sufficient supply of cold water fit for drinking. Where drinking water is obtained from an intermittent public water supply, each workplace shall be provided with storage where drinking water should be stored. Every water supply storage shall be at a distance of not less than



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15 meters from any latrine, drain or other source of pollution. Where water has to be drawn from an existing well, which is within such proximity of latrine, drain or any other source of pollution, the well shall be properly chlorinated before water is drawn for drinking. All such wells shall be entirely closed in and be provided with a trap door which shall be dust and water proof. A reliable pump shall be fitted to each covered well, the trap door shall be kept locked and opened only for cleaning or inspection which shall be done at least once a month.

Washing and Bathing Places:

Adequate washing and bathing places shall be provided separately for men and women. Such places shall be kept in clean and drained conditions.

latrines and urinals:

There shall be provided within the precincts of every workplace, latrines and urinals in an accessible place and the accommodation separately for each of these.

Accommodation for labor:

The Contractor shall during the progress of the Works, provide, erect and maintain necessary temporary living accommodation and ancillary facilities for labour at his own expense and to standard and scales as approved by the Engineer-in-charge. However, following specifications shall be followed:

The minimum height of each hut at the eaves level shall be 2.10m (7ft) and the floor area to be provided will be at the rate of 2.7 Esq. (30sq.ft.) for each member of the worker's family staying with the laborers.

The Contractor shall also construct temporary latrines and urinals for the use of the labourers, each on the scale of not less than four per each one hundred of the total strength. Separate latrines and urinals shall be provided for women.

The Contractor shall construct sufficient number of bathing and washing places, one unit for every 25 persons residing in the camp. These washing and bathing places shall be suitably screened.

15. FIRST AID

First aid box will be kept at project site. First aid trained personnel will be present at the site all the time. First-aid kits or boxes, as appropriate and as per statutory requirements shall be provided at workplaces and be protected against contamination by dust, moisture etc.



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Emergency telephone numbers of nearby hospitals, Police, fire station and administration shall be prominently displayed.

List of Contents for First Aid Boxes

S. No.	Medicine List	PFR Qty	HMEL Keep Qty
1	Small Sterilized Dressing	24	6
2	Medium Sterilized Dressing	12	6
3	Large Sterilized Dressing	12	6
4	Large size Sterilized burns Dressing	12	1
5	Cotton Wool Packets (15cm) Sterilized	12 (15 cms)	1 (75 gms)
6	Savlon 100ml	1	1
7	Beta dine Ointment Tube (Small)	1	1
8	Beta dine 100ml bottle	1	1
9	Scissors	1	1
10	Adhesive Plaster 2x1cm(one roll)	12	1
11	Sterilized Eye pads in separate sealed packets	12	2
12	Copy of First Aid leaflet	1	1
13	Surgical Gloves	0	2
14	Adhesive Band-Aid -water proof	0	10
15	Eye Wash Bottle with distilled water (medium size)	0	1
16	Cifrofloxin / Ciplox Eye Drop.	0	1

Emergency vehicle / ambulance will be kept at the project site round the clock for emergency use.

16. VEHICLE MOVEMENT

Work vehicle shall be as safe stable efficient and roadworthy as private vehicles on public roads. Site management shall ensure that drivers are suitably trained. All vehicle e.g. heavy motor vehicle forklift trucks dump trucks mobile cranes shall ensure that the work equipment



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conforms to the following:

- a. A high level of stability.
- b. A safe means of access/egress.
- c. Suitable and effective service and parking brakes.
- d. Windscreens with wipers and external mirrors giving optimum all round visibility.
- e. Provision of horn, vehicle lights, reflectors, reversing lights, reversing alarms.
- f. Provision of seat belts.
- g. Guards on dangerous parts.
- h. Driver protection - to prevent injury from overturning and from falling objects/materials.
- i. Driver protection from adverse weather.
- j. No vehicle shall be parked below HT/LT power lines.

MAINTENANCE OF VEHICLE:

All Vehicles used for transportation of man and material shall undergo scheduled inspections on frequent intervals to secure safe operation. Such inspections shall be conducted in particular for steering, brakes, lights, horn, doors etc. Driver shall enter in a logbook kept in vehicle and record defect or malfunction appearing in the vehicle during transportation. Site management shall ensure that work equipment is maintained in an efficient, working order and in good repair. Inspections and services carried out at regular intervals of time and or mileage. No maintenance shall be carried below HT/LT power lines.

An equipment passport system (the system certificate to be valid for only three months) shall then be adopted and implemented for each piece of accepted mobile machinery or equipment. The plant and equipment department shall have to implement this equipment passport system inspection. The entries of the inspection carried shall be maintained in a logbook as proof that such inspections are carried out for equipment, which requires annual or third party certification inspection, this passport system should not be treated to supersede such statutory requirement.

DAILY CHECK BY DRIVER:

There should also be daily safety checks containing below mentioned points by the driver before the vehicle is used.



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- Brakes.
- Tires.
- Steering.
- Mirrors.
- Windscreen wipers.
- Wipers/ Fire Protection equipments.
- Warning signals/ Back Horn.
- Specific safety systems i.e. control interlocks.
- Management should ensure drivers carry out these checks.

17. HOUSE KEEPING

HOUSEKEEPING:

Removal of scrap, inflammable material, waste and debris at appropriate intervals. Removal of loose materials, which are not required for use, to be ensured. Accumulation of these at the site can obstruct means of access to and egress from workplaces and passageways.

Workplaces and passageways, that are slippery owing to oil, grease or other causes, should be cleaned up or strewn with sand, sawdust, ash etc.

Proper slogans or signs are used at site

A high degree of housekeeping is to be maintained and the following are to be taken care of:

- All surplus earth and debris are removed/disposed off from the working areas to identified locations.
- Unused/Surplus cables, steel items and steel scrap lying scattered at different places within the working areas are removed to identified locations.
- All wooden scrap, empty wooden cable drums and other combustible packing materials, shall be removed from workplace to identified locations.
- Camp area shall be kept clear and materials like pipes, steel, sand, concrete, chips and bricks, etc. shall not be allowed in the camp to obstruct free movement of men and machinery's.
- Fabricated steel structures, pipes & piping materials shall be stacked properly.



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- No parking of trucks/trolleys, cranes and trailers etc. shall be allowed in the camp, which may obstruct the traffic movement. As well as below LT/HT Power line.
- Utmost care shall be taken to ensure over all cleanliness and proper upkeep of the working areas.
 - Sufficient rubbish bins shall be provided close to workplaces.
 - Bins should be painted yellow and numbered.
 - Sufficient nos. of drip trays shall be provided to collect oil and grease.
 - Sufficient qty. of broomsticks with handle shall be provided.
 - Adequate strength of employees should be deployed to ensure daily monitoring and service for waste management.
 - Different types of rubbish/waste should be collected and stored separately.
 - Paper, oily rags, smoking material, flammable, metal pieces
 - Should be collected in separate bins with close fitting lids.
 - Rubbish should not be left or allow it to accumulate on construction and other work places.
 - Do not burn construction rubbish near working site.

18.EMERGENCY EVACUATION

Respective area emergency coordinator and deputy coordinator shall be appointed to control the emergency situation. Should and emergency required for evacuation, the emergency coordinator or his deputy shall take control of people to observe the following procedure:

In case of an emergency, area supervisor shall notify all employees by using radio on emergency channel or over telephone.

Area emergency coordinator or his deputy coordinator shall decide whether an evacuation is necessary and shall direct the activation of alarm signal accordingly. At the same time emergency controller or his deputy shall be informed about the situation.

Team leader or area in-charge shall direct all employees to assemble in the nearest and designated "Assembly Point" in an orderly manner without making noise.

All plant and equipment shall be switched "OFF" Lighting shall be remained ON

but capable of being switched off shall also be switched "OFF".

Where a work group does not have a dedicated assembly point then team leader shall decide the place where to assemble during time of emergency.

Roll call of the employees shall be taken and status to be reported to the emergency controller.

The area emergency coordinator shall give siren signals for end of emergency period.

Where necessary, in the interests of the health and safety of any person on a construction site, suitable arrangements shall be prepared for dealing with any foreseeable emergency. This



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shall include procedures for any necessary evacuation of the site or any part of it. Step shall be taken to ensure that every person on site knows those procedures and that emergency drills are tested periodically.

A means of detecting and warning of fire must be provided on all sites. Hand bells, whistles, klaxons, manually or electrically operated sounders may be suitable so long as they are clearly audible above background noise in all areas and can be readily identified as being a fire alarm.

EMERGENCY CONTACT NO.

RCM -

HSE -

HR -

FIRE CONTROL ROOM -

OHC -

19. ELECTRICAL HAZARDS IN CONSTRUCTION

Electrical Power is the mainstay of any construction activity. At the same time it requires utmost care in its utilization to avoid accidents due to Electrical Shock, Fire incidents or Electric short circuits. Exposure of electrical installation to adverse environmental conditions increases the risk of such installations to ensure safety of personnel and equipment. This standard address the safety measures required to be adopted for the electrical installations by all contractors during construction phase.

All electrical connections shall be done by an electrician with valid license and to the



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satisfaction of Engineer –In-Charge.

One competent Licensed electrician shall be made available by contractor at site round

The clock to attend to the normal / emergency jobs.

All switch boards / welding machines shall be kept in well ventilated & covered shed.

The shed shall be elevated to avoid water logging. No flammable materials shall be used for constructing the shed. Also flammable materials shall not be stored in and

Around electrical equipment / switch board. Adequate clearances and operational space shall be provided around the equipment.

Fire extinguishers and insulating mats shall be provided in all power distribution centers.

Temporary electrical equipment shall not be employed in hazardous areas without obtaining safety permit.

Proper Housekeeping shall be done around the electrical installations.

All temporary installations shall be tested before energizing, to ensure properEar thing.

All welders shall use hand gloves irrespective of holder voltage.

Multilingual (English, Hindi & Local Language) caution boards, Shock treatment charts and instruction plate containing location of isolation point incoming supply name & telephone no. of contact person in emergency shall be provided in substations and near all distribution boards / local panels.

Operation of earth leakage device shall be checked regularly by temporarily connecting series test lamp (2 Bulbs of equal rating connected in series) between phase and earth.

The following design features shall be ensured for all electrical installations during construction phase.

Each installation shall have a main switch with a protective device, installed in an enclosure adjacent to the metering point. The operating height to the main switch shall not exceed 1.5 M. The main switch shall be connected to the point of supply by mains of armored cable.

The outgoing feeders shall be double or triple pole switches with fuses / MCBs. Loads in a three phase

circuit shall be balanced as far as possible and load on neutral should not exceed 20% of load in



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the phase.

he installation shall be adequately protected against overload, short circuit and earth leakage by the use of suitable protective devices. Fuses wherever used shall be HRC type. Use of re-wearable fuses shall be strictly prohibited. The earth leakage device shall have an operating current not exceeding 30 mA.

All connections to the hand tools / welding receptacles shall be taken through proper switches, sockets & plugs.

All single phase sockets shall be 3 pin type only. All unused sockets shall be provided with socket caps.

Only metallic distribution boxes with double ear thing shall be used at site. No wooden boxes shall be used.

All power cables shall be terminated with compression type cable glands. Tinned copper lugs. Shall be used for multistoried wires / cables.

Cables shall be free from any insulation damage.

Cables shall be laid underground at a minimum depth of 900mm, coverer with sand brick and soil for ensuring mechanical protection. Cables shall not be laid in water logged area as far as practicable. Cable route markers shall be provided at every 25 M of buried trench route when laid above ground; cables shall be properly created or supported on rigid poles of at-least 2 m height. Minimum head clearance of 6 Meters shall be provided at road crossing.

Under ground road crossing for cables shall be avoided to the extent feasible. In any case no underground power cable shall be allowed to cross the roads without pipe sleeve

All cable joints shall be done with proper jointing kit. No. taped / Temporary joints shall be used.

An in depended ear thing facility should preferably be established within the temporary installation premises. All appliances and equipment shall be adequately earthed. In case armored cables are used. The Armour shall be bonded to the ear thing system.

All cables and wire rope used for earth connections shall be terminated through tinned copper lugs.

In case of local ear thing, earth electrodes shall be buried near the supply point & earth continuity wire shall be connected to local earth plate for further distribution to various appliances. All insulated wires for earth connection shall have insulation of green color.



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Separate core shall be provided for neutral earth / structures shall not be used as a neutral in any case.

On / Off position of all switches shall be clearly designated / painted for easy isolation in emergency.

All insulations shall inspected by Engineer-In-Charge at least once in a month

Safety Checklist

- **Welding machine**
- **Grinding machine**
- **Cutting set**
- **Batching plant**
- **Lifting tools/tackles**
- **Bar bending m/c**
- **Bar cutting m/c**
- **All type vehicle**
- **Lathe machine safety checklist**
- **ELCB/RCB checklist**

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RISK ASSESSMENT TASK RECORDS

Prepared By	Reviewed By	Approved By
Safety Officer (ANS)	Safety Manager (ANS)	RCM (ANS)

	Electric Arc Welding
	Gas cutting
	Earth Excavation/RCC / PCC Road Breaking
	Rod cutting, bending, binding and shuttering
	Erection & dismantling of scaffolding
	Lifting, placing and fitting of steel beam on resting cleat
	Confined Space PCC, RCC, Layout, Dressing, Reinforcement Binding, Lifting, Shifting
	Vehicle Entry inside the plant for loading / unloading Soil & Sand & construction materials

TASK: Electric Arc Welding (Pipe line / Steel Structure / Fabrication & erection of MS steel structures)

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS High / Medium / Low	ACTIONS AND ADDL. CONTROLS
1.	Surface preparation (Grinding & 'V' cutting, if required)	1. Flying Particles. 2. Electric shock. Workmen	1. Personal Injury.	1. Ensure ELCB. 2. Metallic body of machine properly earthed. 3. Proper PPE to protect face, eye and hand. 4. Wheel guarding 5. Certified machines	Low	Nil
2.	Welding Job. (Connecting the welding lid & return cable, Insert the Plug of the m/c., Switch on, Insert welding rod, check by striking arc, adjust the welding current, continue welding)	1. Hot surface 2. Electric shock 3. Excessive light 4. Fume 5. Fall from height if work done at height 6. Fire Workmen / Fire in the vicinity	1. Burn Injury 2. Electric shock. 3. Health hazard. 4. Eye irritation. 5. Fire	1. Issue of Work Permit, if working at height / inside confined space. 2. Welding M/c duly certified by Electrical dept. 3. Welding shield with proper shading of glass / Leather hand gloves. 4. Add ventilation / air if work inside vessel 5. Proper platform for working at height & safety belt. 6. Fire extinguisher in the vicinity. 7. Clean surrounding area.	Medium	1. Ensure double earthing of the metallic body of the welding M/c. 2. Ensure the welding return lid is directly connected to the job.

TASK: Gas Cutting

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS High / Medium / Low	ACTIONS AND ADDL. CONTROL S
1.	Cleaning of Cutting Surface	Flying particles <u>Gas Cutter / helper</u>	Foreign body on eye	Gas Cutters' goggles	Low	Nil
2.	Checking of Gas torch & cleaning by pin	Flying particles <u>Gas Cutter / helper</u>	Foreign body on eye	Gas Cutters' goggles	Low	Nil
3.	Opening the gas cylinders	Gas leakage / fire	Injury / property damage	Valve closing Key in place Dual dial Pressure Gauge Valve fitted with Flash Back arrester Gas cylinder secured on trolley	Low	Nil
4.	Adjusting gas pressure at the torch end	Gas leakage / fire	Injury / property damage	Check & clean the torch nozzle	Low	Nil
5	Light up the torch	Fire / burn <u>Cutter / helper / people in the vicinity</u>	Injury / property damage	NRV at torch end Make the surrounding clean No one in front of nozzle near the work area Gas cutter shall wear dark goggles & leather hand gloves	Medium	Appropriat e Gas lighter
6.	Start gas cutting	Fire / burn <u>Cutter / helper / people in the vicinity</u>	Injury / property damage	Clean the surrounding area Gas cutter shall wear dark goggles & leather hand gloves Helper should stand near the cylinder so that in case of emergency, he could close the cylinder		Helper should wear dark goggles & leather hand gloves

TASK: Earth Excavation/RCC / PCC Road Breaking

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS M H L	ACTIONS AND ADDL. CONTROLS
1.	Concrete cutting with hydraulically operated saw	Personal Injury Cutter	1. Cut injury 2. High Noise	1. Rotating part guarding 2. Ear Plug	Low	Nil
2.	Earth cutting (manual excavation)	Personal Injury Sub Contractor men	1. Cut injury 2. Eye injury from flying particles	1. Safety Goggles 2. Spade with good handle 3. Access ladder 4. Area cordoning	Low	Consider manual excavation until utility are identified.
3.	Utility Checking by Authority	Personal Injury	Slip / Fall	Authorized Agency	Low	Nil
4.	Digging with Excavator	Personal Injury	1. Hit by machine during movement 2. Soft soil	1. Signaler 2. Ensure movement on hard soil or provide steel plate 3. Ensure Operator is having skill certificates endorsed by SIL Engineer 4. Equipment Fit Certificate by approved agency 5. Barricade the area	Low	Keep all certificates pasted on cabin glass.

TASK: Rod cutting, bending, binding and shuttering

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS Light / Medium / High	ACTIONS AND ADDL. CONTROLS
1.	Cutting of rods, bending and binding	1. Slip of chisel 2. Slip of hammer from handle. 3. Fly of cutting particles Workmen	Personal Injury.	Use of hammer with proper handle. Use of extended chisel holding tongue. Proper PPE to protect eye and hand.	Low	Nil
2.	Rod bending / cutting using machine	1. Electric Shock 2. Finger injury 3. Catch back Workmen	Personal Injury / OH Hazard	ELCB Proper earthing Hand gloves Keep back straight	Medium	Ensure Emergency Stop Switch is operational
3.	Wooden/ ply shuttering	Nail pocking	Personal Injury	Hand gloves. Remove spent nails	Low	Nil

TASK: Erection & dismantling of scaffolding

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS L M H	Ad. Control
1.	Erection of pipe with shoe, clamps.	1. Slip & fall of pipe & clamps 2. Fall from height of workmen 3. Location hazard Workmen	1. Personal Injury. 2. Equipment damaged.	1. Issue of WORK 2. Constant Supervision. 3. Loose materials like pipe clamps tools to be tied up with ropes 4. Use of Safety belts with nearby permanent structure. 5. Surrounding area to be cleaned & cordoned 6. Vertical life line & fall arrester to be laid, where required. 7. Area cordoning	M	1. Use full body safety belt with double clamp. 2. Proper strength for load bearing scaffolding to be considered. 3. Follow standard scaffolding design 4. Get the Scaffolding certified by Competent Person
2.	Staging / working platform	--DO--	--DO--	In addition, 1. Arrangement for lifting the steel staging plates / wooden planks of min. 1 ½ inch thick. 2. Approach ladder	M	Same as above
3.	Dismantling of Tubular scaffolding	1. Slip & fall of pipe & clamps 2. Fall from height of workmen 3. Location hazard	1. Personal Injury 2. Equipment damaged	1. Issue of WORK PERMIT 2. Constant supervision. 3. Loose materials like pipe clamps tools to be tied up with ropes 4. Use of Safety belts with nearby permanent structure. 5. Surrounding area to be cleaned & cordoned 6. Vertical life line & fall arrester to be laid.	M	Same as above

				Use of hand gloves.		
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TASK : Lifting, placing and fitting of steel beam on resting cleat

Sl.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS L M H	ACTIONS AND ADDL. CONTROLS
1.	Positioning of beams on the resting cleat.	1. Fall from height 2. Rubbing with hand. Contractors' employee	1. Loss of life 2. Palm injury	1. Obtain Permit To Work 2. Use hand gloves 2. Steel ladder erected along with the column. / Use certified lifting chairs & crane to lift the workmen 3. Clamping of Fall Arrester with life line (attached with the full body Safety Belt of the concern person) with the vertically laid life line. 4. Working platform may be erected at height. Check the condition of welding & appropriate support.	M	1. Close Supervision for strict adherence with the existing controls. 2. Make the area out of bounds for others
2.	Matching of holes of the beam with the cleat	---- DO ----	---- DO ---	---- DO ----	M	Close Supervision for strict adherence with the existing controls.
3.	Use of gas cutter (if req.) for matching with the hole. (For doing the job listed in step 3-5, the concerned person has to sit on floor beams away from vertically laid lifeline.)	1. Burn Injury 2. Fall from height 3. Rubbing with hand. Contractors' employee	---- DO ---	1. Use hand Gloves. 2. Steel ladder erected along with the column. 3. Clamping of Fall Arrester (attached with the full body Safety Belt of the concern person) with the vertically laid life line / Use certified lifting chairs & crane to lift the workmen. 4. An addl. rope (clamped to the safety belt) other than fall arrester to be anchored with the beam while stay away from the vertical	M	Close Supervision for strict adherence with the existing controls.

				lifeline. 5. Gas Cutters goggles. 6. Regular checking of gas cutting torch and rubber tubing. 7. Ensure Flash Back arrester 8. Securing of gas torch at height.		
4.	Fixing of fasteners and tightening.	1. Fall from height 2. Rubbing with hand. Contractors' employee	---- DO ---	1. Hand Gloves. 2. Steel ladder erected along with the column. 3. Clamping of Fall Arrester (attached with the full body Safety Belt of the concern person) with the vertically laid life line. 4. Working platform may be erected at height. Check the condition of welding & appropriate support.	M	Close Supervision for strict adherence with the existing controls.
5.	Welding job at height	1. Burn Injury 2. Fall from height 3. Rubbing with hand. Contractors' employee	---- DO ---	1. Safety Helmet and shoes, Gloves. 2. Steel ladder erected along with the column. 3. Clamping of Fall Arrester (attached with the full body Safety Belt of the concern person) with the vertically laid life line. 4. An addl. rope (clamped to the safety belt) other than the fall arrester to be anchored with the beam while stay away from the vertical lifeline. 5. Head mounted Welders' Screen. 6. Regular checking of Elec. holder, cable and welding set. 7. Securing of electrode holder at height.	M	Close Supervision for strict adherence with the existing controls.

6.	Climbing or descending from height	1. Fall from height Contractors' employee	1. Loss of life	1. Safety Helmet and shoes, Gloves. 2. Steel ladder erected along with the column / Use certified lifting chairs & crane to lift the workmen 3. Clamping of Fall Arrester (attached with the full body Safety Belt of the concern person) with the vertically laid life line.	M	Close Supervision for strict adherence with the existing controls.
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TASK-Confined Space PCC, RCC, Layout, Dressing, Reinforcement Binding, Lifting, Shifting

SL.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS H M L	ACTIONS AND ADDL. CONTROLS
1.	Site Preparation	Damages due to use of defective tools and lack of job clarity	Ensure that tool box talk is being carried out about the execution of the jobs and about right tools by the contractor superiors	M	Use vertical life line and Fall Arrester along with Full body Safety belt during climbing.
2.	confined Space RCC, PCC Bar Binding	Injury to Persons/Asset	- Ensure proper communication system between stand by person and working person inside the pit. - Ensure entry and exit record of person	M	Not required.
3.		Fire due to sparks	Inspected ladder by safety dept. must be used use gum boot and rubber gloves.	M	Tie the tools with rope to prevent fall from height.
		Injury to Persons/Asset	Ensure that use of mandatory PPEs use	M	

			dust mask.		
			Ensure that fire extinguisher available before starting the job	M	
			Ensure that GAS test to be done in every 2hours	M	
			Area to be Hard Barricated	M	
SL.	SEQUENCE OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS H M L	ACTIONS AND ADDL. CONTROLS
	Shuttering/Rebar Material Bar Binding	Injury to Person	Ensure all tools shall be in good condition Ensure proper stacking of material at designated location	M	
	Preparation & Checking	Fire explosion due to spark generated	Spark Arrestor must be healthy.	M	
			Check LEL inproposed route of vehicle.	M	
			Switch off engine and evacuate if F & G Alarm is sounded.	M	
	Entry Inside Plant	Damage to work place due to hitting against pipe/equipment	Vehicle will not enter paved area of plant; if such an entry is required, necessary permission from plant manager/production manager will be necessary. Identify all pits near to the route, barricade/highlight if required.	M	
		Injury to people around	Do not use bear handed to handle heavy material.	M	
	Loading / Unloading of material	Injury to people during material handling	Do not drop material from height. Use correct lifting/lowering devices.	M	

TASK-Vehicle Entry inside the plant for loading / unloading Soil & Sand & construction materials



ANS CONSTRUCTIONS PVT. LTD.

RISK ASSESSMENT PROCEDURE

Prepared By	Reviewed By	Approved By
Safety Officer (ANS)	Safety Manager (ANS)	RCM (ANS)

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PROCEDURE FOR RISK ASSESSMENT

Introduction:

Risk assessment is a careful examination of the job steps & its work place to identify hazards and how it could injure people or cause loss of property. On completion of the risk assessment, recommendations are made so that enough precautions are taken during the job execution to prevent harm and injury and bring down the risk to tolerable level.

Definitions:

Hazard: It means any thing, which can cause harm (e.g. Chemicals, electricity, working from ladder etc.)

Risk: It is a combination of the likelihood and consequence(s) of a specified hazardous event occurring.

At ANSCPL, Bathinda we need to evaluate risk of following jobs to decide whether existing precautions are adequate or additional precautions are required to ensure that risk is brought down to tolerable level

- All Civil and Mechanical Jobs,
- Process safety.
- Any new activity related to operation.
- Mechanical material handling activity.
- Manual material handling activity.
- Confined Space
- Excavation

Procedure:

Assessment process

- 1. Listing of jobs:** Prepare the list of jobs, which come under purview of above mentioned criteria. Job shall be categorized for making risk assessment ***e.g. example Impeller changing of all pumps in corrosive service shall come in one category and Impeller changing of flammable chemical will come in another category. There is no need of making equipment wise risk assessment***

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2. Breaking job into sub activity: Each job identified for the risk assessment shall be broken down into sub activities.

3. Look for the hazard: Visualize hazard in each sub activity. Walk around the workplace look afresh to assess what could reasonably be expected to cause harm.

4. Decide who might be harmed and How.

5. Decide what precautions are needed to eliminate the risk or bring it down to acceptable level.

6. Record of all Risk assessment shall be in the **attached** format

Criteria for risk decision:

Based on the risk score the risk will be decided as trivial, tolerable moderate, substantial or intolerable. Risk score will be on the basis of risk rating. Risk rating will be on the judgment of risk assessment team

Following index shall be used as guideline for risk decision:

Severity rating	Fatal*	5	10	15	20	25
	Extremely harmful	4	8	12	16	20
	Harmful	3	6	9	12	15
	Slightly Harmful	2	4	6	8	10
	No Effect/ Trivial	1	2	3	4	5
		Almost impossible	Very unlikely	Likely	Probable	Certain
Likely hood rating						

Guide line for deciding severity rating:

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Fatal: - Where death is imminent or total permanent disability can take place. There is total loss to the plant
Extremely harmful: - Where partial permanent disablement can take place Or there could be irreversible health effect. Amputation, major fractures, poisoning, occupational cancer, other severely life-threatening diseases can occur. For property damage it is more than 5 lac rupees.
Harmful: - Lacerations, burns, contusion, serious sprains, minor fractures, dermatitis, asthma, work related limb disorders, Ill health leading to permanent minor disability. For property damage it is more than one lac but less than 5 lacs.
Slightly Harmful: -Superficial injuries, minor cuts, eye irritation temporary discomfort. For property damage it is more than 10 thousand but less than one lac.
No Effect / Trivial

Guide line for deciding likely hood rating:

Almost impossible: -
Very unlikely: - Never happened till date, But still felt that it could be possible.
Likely: - Infrequent impact (Typically once in a year). Has been observed in similar circumstances. 10% to 50 % chances of occurrence.
Probable: Repeated impact (at least once in a quarter) More than 50 % chance of occurrence.
Certain: Accident is sure to occur.

Risk ranking and action:

Risk score		Risk Level		Action	
1-2		Trivial		No action	
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3-6	Tolerable	Instructions/ Toolbox talk
8-10	Moderate	Instructions /Toolbox talk/ Training to reduce the risk level to tolerable label.
12-16	Substantial	Training & additional measures or Objectives and Programs to be formulated to reduce the risk level to tolerable level.
20-25	Intolerable	Re- Risk assessment & additional control measures or Objectives and Programs to be formulated. Work should not be started unless the risk level has been reduced to tolerable level

Risk assessment team:

- Plant safety representative (PSR), the concerned activity supervisors and operator or the technician or contractor's representative as the case may be shall be the team member for risk assessment. During silent hours if any job is to be taken up for which risk assessment has not been carried out previously the concerned officers shall prepare it.
- If a new chemical is used or new equipment is used for the first time or any activity which is one-time job and which will continue for more than one day safety officer shall also be involved in the risk assessment. If PSR and safety officer need the representative of Plant Maintenance, ESD will join as team member.
- External experts shall also be involved based on the job requirement e.g. material handling of heavy equipment.
- Contractor supervisor (in charge of the activity) shall also be a team member if the job is to be done involving contract employees.

Field application:

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- Outcome of the risk assessment shall be used to prepare Occupational Health & Safety objectives & targets.
- Current approved risk assessment for the job is to be attached with the work permit.
- The serial No. of the risk assessment shall be written on the permit at the appropriate place provided in the work permit.
- The job executing supervisor shall ***explain the risk associated with the activity and ensure that required precautions are taken as per the risk assessment.***

Reviews:

- The safety section head shall review all risk assessments where safety officer is involved.
- All risk assessments made first time shall be reviewed by concerned plant HOD.
- All the risk assessment shall be reviewed once in five years by the team and concerned HOD and revised if needed.
- If any accident takes place the risk assessment shall be reviewed for that job.

Training:

The supervisor of the concerned job shall impart training based on the risk assessment to the person / persons doing the job.

Record:

All the risk assessment sheets shall be Indexed and kept in file by PSR and a copy shall be available in the department & control room for ready reference.

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Risk Assessment Serial no:
Checklist No:

RISK ANALYSIS

Name of the equipment/location:

Activities:

Purpose:

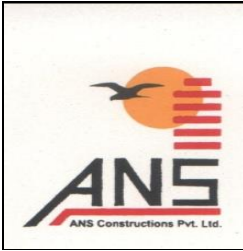
Brief description of activity:

Sub Activities:

- 1
- 2
- 3.

SI.	SEQUENC E OF BASIC STEPS Evaluated as Critical	HAZARDS & PERSONS AT RISK	POTENTIAL HARM	EXISTING CONTROLS & SAFE PROCEDURES	EVALUATION OF RISKS High / Medium / Low	ACTIONS AND ADDL. CONTROLS

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EMERGENCY RESCUE PLAN

- Ensure that at least one competent person shall always available with confined entry job.
- The attendant will maintain IN & OUT register for recording attendant entrant during entry & exit from the confined space.
- In case of emergency the attendant will immediate report to project through walky talky or IS phone on given number -----,
- The attendant will also report to fire control room through walky talky or call on – ----- and give the clear message of incident and proper location of the incident.
- No any other person will move inside the confined area to take out the person expect rescue team. This will be ensured by supervisor & safety officer of M/s ANS CONSTRUCTIONS PVT.LTD.
- Give proper First Aid treatment at site to the workman, if required.
- In case of critical situation, rescue the person and send to OHC by Vehicle/ Ambulance for further treatment.
- Other workers who are working nearby area will follow the emergency escape route and will be assemble at the assembly point.
- Rescue team (HMEL-Fire department) will be reach the location within minute and will take necessary action.



ANS Constructions Private Limited

HEALTH, SAFETY & ENVIRONMENT (HSE) POLICY

ANS CONSTRUCTIONS PVT.LTD. (ANSCPL) is committed to comply with the requirement of HSE Management System & continually improve its effectiveness by providing the appropriate processes, resources, principles, training & work culture.

In this endeavor, ANSCPL shall strive to be conscious of its responsibility towards creating, conserving & ascertaining a clean environment for sustainable development and protect the health & safety of all its associated workforce, ensuring thereby a safe, healthy & accident free work place.

To meet this commitment towards achieving operational excellence in HSE objectives; ANSCPL shall:

- Adopt appropriate operational practices, safe system & suitable methods to monitor, control & minimize the risk & impact of the activities on HSE.
- Ensure that its operations at least comply with applicable HSE laws, regulations & statutory provisions.
- Ascertain that appropriate resources are made available to implement HSE policy & continuously review the policy 's relevance with respect to legal, community & business development.
- Develop & maintain a motivated workforce trained for effective management of HSE emergency situations.
- Provide information on HSE policy to the concerned authorities & interested parties and conserve with business associates to facilitate their health and safety performance improvement with the intent of their individual HSE obligations.
- Cultivate a work culture where the HSE policy is understood, implemented & maintained by employees at levels within the organization.
- Continually improve its HSE performance by setting objectives & targets and conduct regular audits to ensure that operations are in compliance with HSE management system requirements.

HARDEEP SINGH

CHIEF EXECUTIVE OFFICER



ANS CONSTRUCTIONS PVT. LTD.

QUALITY POLICY

The quality management policy of ANS CONSTRUCTIONS PVT.LTD is to established, implement, promote and maintain a quality management system to achieve the following goal.

- Obtain customer satisfaction and confidence by deploying facilities, systems and services, which shall confirm to customers requirements.
- Conformance to applicable statutory regulations and Indian standard on quality system.
- Continuous improvement in performance and development of staff for quality management activities.

The above goods are achieved by participating of all concerned -----

We will achieve above goals by particulars of all concerned employees in quality managements system activities.

The quality management system activities are established & are implemented & maintained with company wide effort & harmony.

The company evaluate the performance of quality management system activities objectively in management review meeting.



This policy shall be reviewed at defined intervals to ensure its suitability and effectiveness.

ANS CONSTRUCTIONS PVT. LTD.

QUALITY ASSURANCE STANDARDS

ANSCPL will deploy a qualified and experienced quality assurance Engineer at the site. The QAE will be responsible for formulating a project- specific quality plan and its implementation.

The quality plan will enumerate the documented procedures for the execution of the various activities as included in the scope of works. The quality plan will broadly include procedures for planning and monitoring, drawing control, incoming materials inspection, receipt, acceptance and testing, procedure for calibration of measuring of test equipments, Procedure for selection and evaluation of sub contractor's vendors, procedure for corrective and preventive action etc.

The quality plan will clearly specify the acceptance criteria for quality affecting material acceptance criteria for various quality affecting activities handling over procedures for complaint handling and the procedure for corrective preventive action. It is mandatory for ANSCPL staff to study the quality plan in details and implements the same in its totality every staff member is considered as a quality head and is made responsible for the area of his jurisdiction.



The quality control Engineer will be the authority to monitor the inspection and tests of the materials and the activities. The inspections records will be stored in a controlled location to aid easy retrieval and tractability.

The ANSCPL site management will take steps to inculcate quality awareness amongst the constructions Engineers and will educate them W.R.T. the contract requirements, constructions program, finish quality requirements, co-ordination requirements and the over all expectations of the clients. ANSCPL project manager will personally manage the quality assurance aspects of the project and will be assisted by the team of planning Engineer, the QCE and the construction Engineer. Weekly toolbox meetings will be conducted where on only quality matters will be taken up for discussions. Other aspects like safety and house keeping will also be conducted by the QCE.

The site constructions engineer directly executing the job shall inspect the job done as per the documented procedures and the checklist for the given job. The occurrence of any non-confirmative shall be recorded and corrective actions taken. The QCE will also check and ensure compliance to the specified requirements. The results and the conclusions of such inspection shall be documented through quality records. These records shall be maintained at the quality control department. The documents shall be codified and indexed to aid easy retrieval and ensure tractability.



The closure of all non-conformities and the corrective actions shall be carefully monitored by the project manager of ANSCPL. The non-conformity report shall form an important input to the management review meetings.

On substantial completion of the works ANSCPL QCE and the planning Engineer shall inspect all works specified in the contract and look for any non conformity vis- a- vis the constructions drawings and the technical specifications. A detailed list will be prepared by ANSCPL enlisting those areas where actions to be taken to bring the works in line with the contract requirements.

A task force will be setup to attend to these areas where actions need to be taken. This task force will be resourced with skilled workmen. Regular monitoring will be done to ensure gradual minimization in the number of non-conformity. Such corrective actions will be documented and the approval of the executed jobs will be taken from the clients QCE.

ANSCPL believes that quality is built in not inspected in hence ANSCPL will ensure that even if the clients QCE is not available the client is guaranteed a quality product.

ANSCPL have a zero-defect mentality and our systems back this up. ANSCPL believes in the tackling problems as they occur and not passing defective work down the line. ANSCPL will take care to leave no unattended points in the subtasks that finally go in to make the end product. This will be ensured with



the use of QA sheets for respective activities. As a result, no defective works is allowed to be passed down the line. The idea of QMS is to avoid totally any reworks expenses.

ANS CONSTRUCTIONS PVT. LTD.

QUALITY IN PROCUREMENT

- 1- Documented procedures shall be established and maintained to ensure that purchased product / service conforms to specified requirements.
- 2- The procurement process starts with the identifying and selections of vendors on. The basis of their capability to supply materials and services as per the specified requirements. The vendors so selected will be enlisted as per the documented procedure.
- 3- Vendor's performance in items of delivery and quality will be continuously evaluated as per vendor's performance evaluation procedure.



Extent of control shall be primarily based on the records of the past performance and where applicable, the criticality of the product or service provided.

- 4-Quality records of acceptable sub-contractors shall be maintained (Refer procedure for control of quality records)
- 5-All materials shall be purchased as per the documented procedure as may be necessary for the project from the approved vendors. The materials received at the site shall be inspected or otherwise verified as per the documented procedure to ensure that they conform to specified requirements records of such inspection and testing shall be maintained at the project location. Such materials shall be handled and stored as per the documented procedure to prevent damage/ deterioration. All incoming materials shall be identified with regards to their status of inspections and testing. The identification and inspection and test status shall be by labels, containers, locations, and boards as per the documented procedure.
- 6-Specifications of all the materials shall be clearly described in the indent and subsequently in the purchase order and these shall conform to the specific requirements of the customers as mentioned in the contract. All purchase orders before release to the vendors shall be reviewed for adequacy and completeness of requirements



and approved by the approving authority as per the documented procedure.

- 7-Procurement of capital items like plant, machinery, vehicles, etc. shall be governed by the relevant documented procedure.
- 8-Where it is proposed to verified the purchased products at the subcontractor's premises, verification details and methods of product release shall be details in the purchase order. If specified in the contract the customer/ representative shall be provided all facilities to carry out inspections of purchased product at the suppliers/ subcontractor's premises.
- 9-The products supplied by customer for the project shall be verified, stored and maintained as per the documented procedure. Any such product that is not acceptable shall be reported to the customer Records of such product shall be maintained.
- 10- Any non-conformity regarding incoming / purchased materials shall be reviewed and disposed as per documented procedure. Responsibility for review and authority for the disposition of the non-confirming incoming / purchased product shall be defined. Where materials are accepted with concession / rework / repair the same is done with the concurrence of



the clients/ client representative. Such non-conformities that have been accepted / repaired shall be recorded to denote the actual condition. Repaired / Rework product shall be re- inspected as per documented procedures.

ANS CONSTRUCTIONS PVT. LTD

INSPECTION AND TEST PLAN

Inspection and Test Plan (ITP) are prepared to exercise stage wise control during various phases of construction i.e. before during and after construction. The plan indicates the stages upto



which work can proceed with their own quality control and stages where specific approval of consultant/client is required. The plan specifies if the test is to be witnessed at Random or not and the documents to be maintained.

Control of Non-Conforming Product

Any works at construction site that does not conform to specific requirement shall be identified to prevent use, dispatch or mixing with conforming products. The relevant function shall be notified and work segregated from conforming work pending evaluation and disposition.

The decision to rework, regarded or accept with or without repair by concession (waiver to specified requirement) shall be made by the appropriate authorities i.e. design engineer, certifying authority. Records shall be maintained for reworked or regarded acceptance with remarks etc. Non-conforming report will be prepared & resolution passed. The cause of non-conformation shall be investigated and corrective action taken.

Various formats have been deployed to have quality control at various stages of construction as per requirements of clients.